

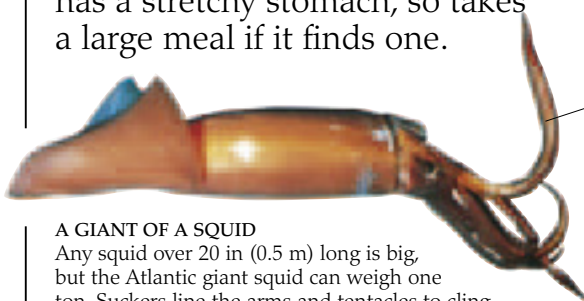
The twilight zone

BETWEEN THE BRIGHT SUNLIT WATERS of the upper ocean and the pitch black depths is the half light of the twilight zone, which ranges from 650 to 3,300 ft (200 to 1,000 m) below the surface. Fish living in the twilight zone often have rows of light organs on their undersides to help camouflage them against the little light filtering down from above. These glowing lights can be produced by chemical reactions or by colonies of bacteria living in the light organs. Many animals, including some lantern fish and a variety of squid, spend only their days in the twilight zone. At night they journey upward to feed in the food-rich surface water. By doing this, they are less at risk from daytime hunters such as seabirds. Others, such as the lancet fish, spend most of their lives in the twilight zone eating any available food. The skinny lancet fish has a stretchy stomach, so takes a large meal if it finds one.



HUNTER OF THE DEPTHS

Viper fish have an impressive set of daggerlike teeth to grab their fish prey, which they attract with a lure dangling from the front of the dorsal fin. The extralong teeth in the bottom jaw are too large to fit inside the mouth when the jaws are closed. When swallowing prey, such as a hatchet fish (above left), the hinged jaws open very wide.



Jumbo squid can reach 12 ft (3.6 m) to the tips of their tentacles

A GIANT OF A SQUID

Any squid over 20 in (0.5 m) long is big, but the Atlantic giant squid can weigh one ton. Suckers line the arms and tentacles to cling onto prey. Sperm whales often bear sucker-shaped scars where they have grappled with squid (p.34).



MERMAN

Many strange creatures lurk in the ocean depths, but no one is likely to find one looking like this.

Dorsal fin can be used for herding fish prey

Model of a lancet fish

Large gill flap

Pointed teeth for grabbing fish

Pectoral fin

LONG AND SKINNY

The lancet fish only weighs up to 10 lb (4.5 kg), because it has a narrow body, lightweight bones, and not much muscle. A predator, it catches squid and other fish, such as hatchet fish, living at the same depths.

Pelvic fin

Fin ray

Saillike dorsal fin can be raised and lowered